

GenAI Customer Service Email Generator

Sprint 1 Review

Team Members: Sarah Kim, Michael Chen, Jessica Rodriguez, David Thompson

DSS5105 Capstone Project • AY2026/27 • 5-min live presentation + 2-min Q&A

Problem, Solution & Business Value

Key Problems

- **60-70% of customer service time** spent writing repetitive emails
- **24-48 hour response delays** affecting customer satisfaction
- **Inconsistent communication quality** across service representatives

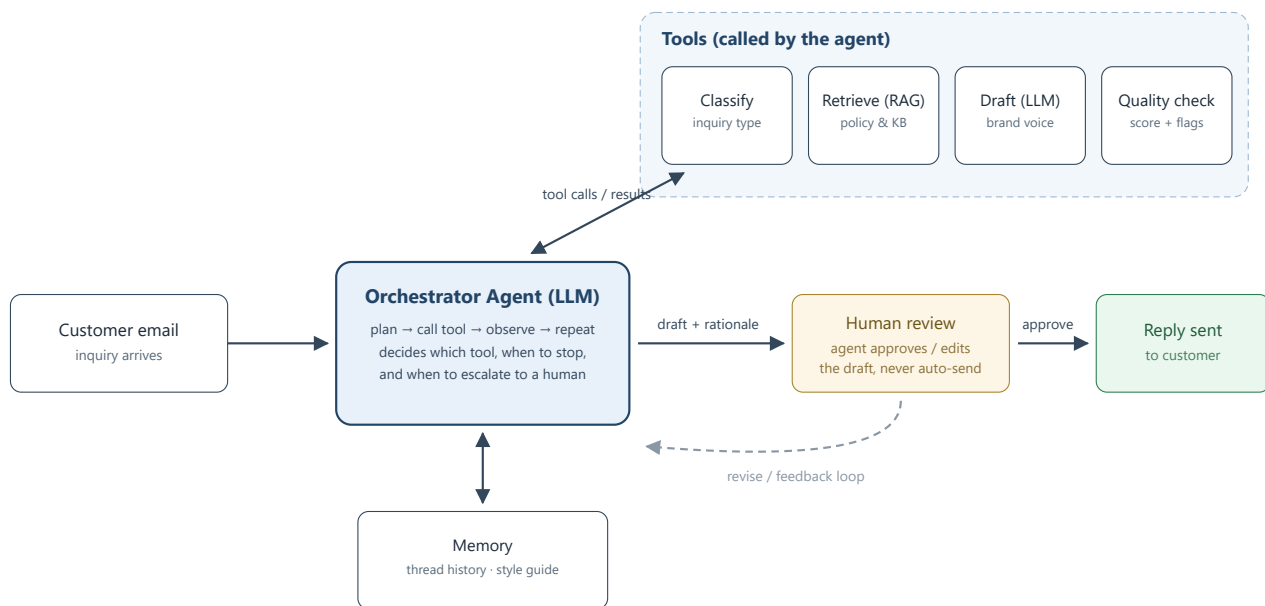
GenAI-Powered Solution

Automated email generation with human-in-the-loop workflow — LLMs draft emails, customer service teams review and approve before sending

Business Value

- **60-70% faster response times** through automated drafting
 - **Increased agent productivity** enabling focus on complex issues
 - **Consistent brand communication** across all customer interactions
-

System Architecture — Agentic LLM



Sprint 1 Progress (Optional)

Data Foundation & Analysis

- **Email corpus collection** — 10K+ customer service email examples
- **Inquiry classification system** — Categorized common customer request types
- **Quality criteria definition** — Established evaluation metrics for generated emails

Technical Infrastructure

- **Development environment** setup with Python/LangChain ecosystem and version control
 - **Initial GenAI prototypes** — GPT-based email generation baseline model
 - **Human-in-the-loop workflow** — Vetting interface and approval process design
-

Risk Assessment

#	Risk	Probability	Impact	Mitigation
1	Generated emails contain inappropriate or inaccurate content	High	High	Mandatory human vetting workflow and quality scoring system
2	GenAI model fails to maintain consistent brand voice	Medium	High	Fine-tuning with company email corpus and style guidelines
3	Customer service team resistance to AI-generated content	Medium	High	Change management program and gradual rollout with training

Key Assumptions

#	Assumption	Validation Required	Risk Level	Dependency
1	Email corpus quality is sufficient for training effective prompts	Data audit completion	Medium	Customer service team data access
2	GenAI models can maintain appropriate tone and accuracy	Model validation testing	High	Prompt engineering effectiveness
3	90% human approval rate is achievable for generated emails	Baseline prototype results	High	Quality framework success

Sprint 2 - Development Roadmap

Development Area	Priority	Impact	Description
Basic GenAI email generation	High	High	Simple prompt-based draft creation for common inquiries
Human review workflow	High	High	Basic approval interface for customer service team
System integration	High	Medium	Connect email input to GenAI output with manual review step
Initial testing	Medium	Medium	Validate core functionality with sample customer emails

Sprint 2 Success Criteria

Focus: Basic system that works — functional prototype rather than optimization

Core Deliverables

- **Core email generation working** — System can create basic drafts for routine inquiries

- **Human review process functional** — Team can approve/reject generated emails
- **End-to-end workflow complete** — Inquiry input to customer email delivery

Acceptance Criteria

- Generate emails for at least 5 common inquiry types
- Achieve 70%+ approval rate in initial testing
- Complete processing within 30 seconds per email